

ARCHITECTURE REVIEW CHECKLIST

Mandatory review before implementation

Document	Version	Status	Language
ES-03	1.1	FROZEN	English

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Frozen Backbone

- ☐ Package responsibilities are preserved.
- ☐ Dependency direction is correct.
- ☐ Control flow is correct.
- ☐ Observation flow is correct.

Composition

- ☐ RobotContainer only creates objects, injects dependencies, and configures bindings.
- ☐ No mechanism logic is placed in RobotContainer.

Commands / Subsystems

- ☐ Commands do not call vendor APIs.
- ☐ Subsystems own mechanism state and logic.
- ☐ Command requirements are declared correctly.

IO

- ☐ Each mechanism has its own IO interface.
- ☐ Each IO interface owns its own Inputs snapshot.
- ☐ The IO implementation populates Inputs.
- ☐ IO does not publish NetworkTables.

Observation

- ☐ Evaluators are pure, stateless, deterministic, and side-effect free.
- ☐ Observation does not access hardware, vendor APIs, NetworkTables, CommandScheduler, RobotContainer, mutable mechanism state, or control behavior.
- ☐ Observation is a vendor-neutral, read-only model.
- ☐ A subsystem or estimator produces an immutable Observation from IOInputs.
- ☐ frc.robot.observation is recognized as a permanent top-level package.

Telemetry

- ☐ Telemetry is observer-only.
- ☐ Telemetry does not schedule commands.
- ☐ Telemetry does not read hardware directly.
- ☐ Publishers use explicit types and stable topics.
- ☐ Telemetry only consumes and publishes Observations; it does not mutate an Observation or control behavior.
- ☐ util remains generic; mechanism-specific observation meaning is not placed in util.

Quality

- ☐ No duplicated responsibility.

- ☐ No over-engineering.
- ☐ No magic numbers.
- ☐ A build and verification plan exists.

Review Result

Module	
Reviewer	
Result	PASS / FAIL
Notes	

Revision History

Version	Date	Status	Notes
1.0	2026-07-18	FROZEN	Initial release
1.1	2026-08-01	FROZEN	APPROVED: add permanent observation boundary; control flow unchanged.